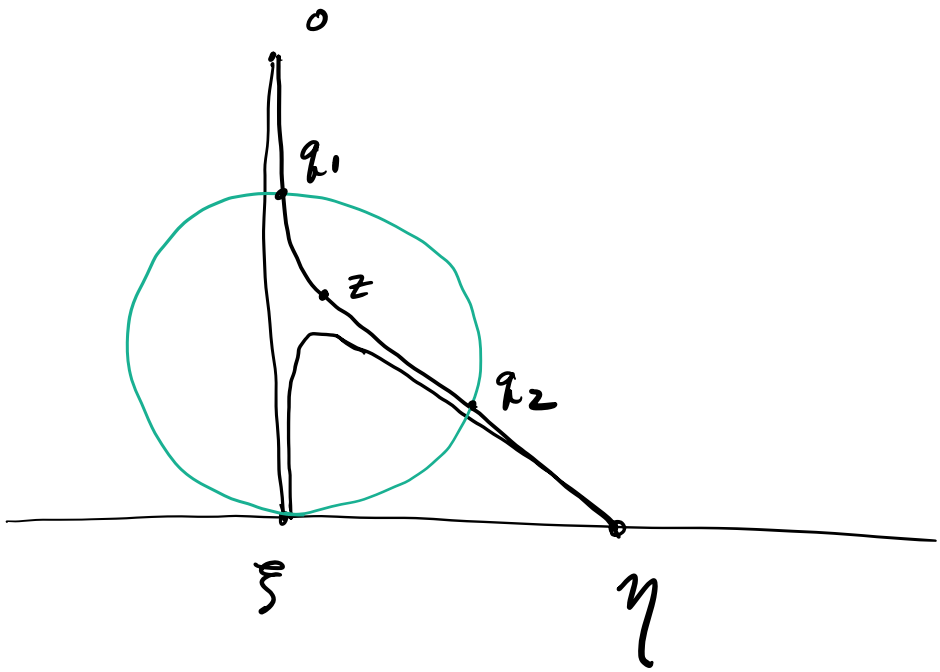
$2 \times$ maximal depth in $H \approx$ total time in H

<u>Pf idea</u>



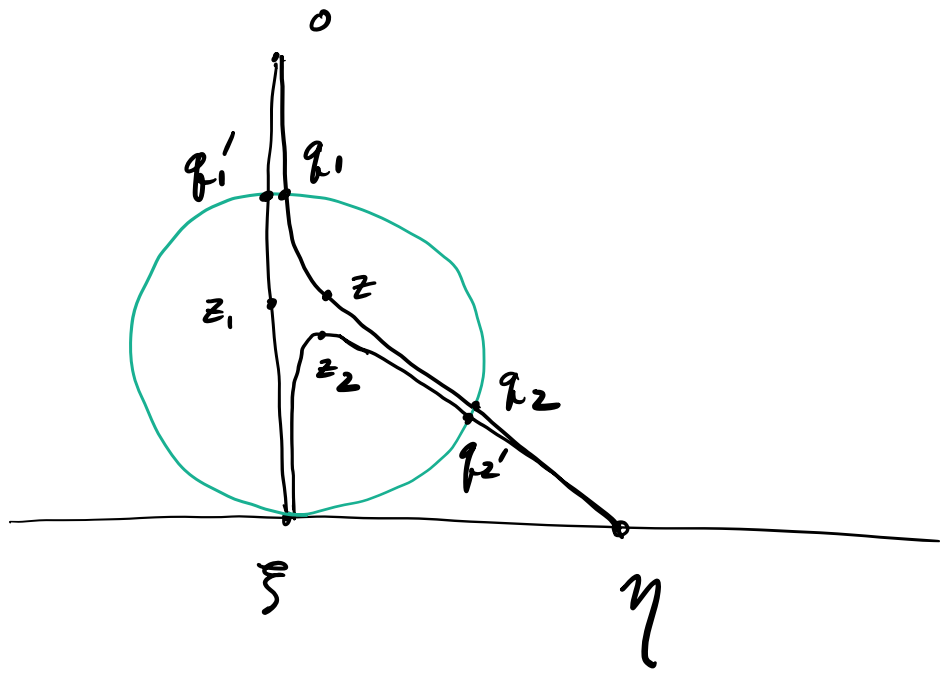
Let $z \in \mathrm{proj}_{[0,\eta)}(\xi)$. Then:

- z bndd dist. from inner Δ of $0, \xi, \eta$
- z minimizes $\beta_\xi(x, 0)$ for $x \in [0, \eta)$
- z is a point of tangency of $[0, \eta]$ wrt horoballs centered at ξ.

- <u>In particular</u>,

$d(q_1, z) \approx d(z, \Gamma_0) \approx$ max depth of $[0, \eta)$ in H

WTS: $d(q_1, z) = \dfrac{d(q_1, q_2)}{2}$



see that for $i = 1, 2$,

$$d(q_i, z) \approx d(q_i', z_i) \approx \beta_\xi(q_i, z)$$

Since $\beta_\xi(q_1, z) \approx \beta_\xi(q_2, z)$, we have

$$d(q_1, z) \approx d(q_2, z).$$

Then $d(q_1, q_2) = d(q_1, z) + d(z, q_2)$

$$= 2\, d(q_1, z) \quad \square$$